

Mohamed Amine Jmal

AI Backend Engineer

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SUMMARY

Backend engineer with ~3 years of experience shipping production systems — including a 21-table ERP in live use and a LangGraph-based pipeline processing ~200 tenders/week — now specializing in RAG and agentic AI (LangGraph, RAG evaluation, vector search) for backend infrastructure.

EXPERIENCE

AI Engineer Intern — Zetabox Jun 2026 – Aug 2026

Sfax, Tunisia

- Built and delivered an internal tender-intelligence pipeline that scrapes TUNEPS (Tunisia's national public tenders platform) and routes each tender through a LangGraph state graph — with nodes for relevance ranking, detail-page augmentation, and proposal drafting — now used operationally within the team, processing ~200 tenders/week.
- Chose a single modular FastAPI service over microservices to prioritize shipping speed for an early-stage tool, and automated the full scrape → rank → draft flow with APScheduler on a recurring schedule.
- Designed the full PostgreSQL data layer from scratch: modeled the schema in DBML, implemented it with SQLAlchemy async ORM, and managed schema evolution with Alembic async migrations.
- Evaluated GraphRAG architectures (Microsoft GraphRAG, LightRAG, Fast GraphRAG) and presented a comparative analysis to the engineering team.

Software Engineer — Freelance Mar 2025 – Present

Bun, Hono, Drizzle ORM, PostgreSQL, JWT, Docker, Shadcn, Next.js

- Built and deployed a 21-table PostgreSQL-backed ERP and payroll automation system for a live industrial-sector client — admin, inventory, sales, HR, finance, commercial, and marketing modules — now in production for ~20 employees, with role-based access control and JWT authentication across all 7 modules.
- Set up CI/CD with GitHub Actions and containerized the stack with Docker Compose for consistent, repeatable deployments.

PROJECTS

EcoShield – Water Monitoring System

Express, FastAPI, RabbitMQ, Redis, MongoDB, SSE, JWT, Docker, LSTM, Next.js

- 1st place, AI Night Challenge hackathon; later extended into a production-oriented architecture.
- Real-time sensor pipeline using RabbitMQ topic exchanges, time-bucketed Redis keys with last-known-value imputation, and per-sensor JWT auth with encrypted keys in MongoDB across multiple locations; two LSTM models handle flow forecasting and false-data anomaly detection.

RAGwise – RAG Evaluation Pipeline (Ongoing)

Docling, BGE-M3, Qdrant, LangGraph, RAGAS, Langfuse, Cohere Rerank, FastAPI

- Built an evaluation-driven RAG pipeline over ~3,069 arXiv papers: parsed with Docling, embedded with BGE-M3, indexed in Qdrant, orchestrated with LangGraph.
- Used RAGAS to build a knowledge graph from 200 document chunks (200 nodes, 11K+ relations) and auto-generate a 100-question testset with expected answers and source references; tracked evaluation runs with Langfuse.
- Benchmarked a hybrid dense+sparse retrieval baseline (relevance 0.73, precision 0.45, recall 0.67, faithfulness 0.92, 1.65s latency) against a Cohere-reranked variant that lifted precision 33% (→ 0.60), recall 7% (→ 0.72), and faithfulness 4% (→ 0.96), holding relevance flat and trading off latency (→ 3.22s).
- Diagnosed and fixed a bug in Docling's HTML backend that silently dropped <table> elements nested inside <figure> tags (common in LaTeXXML/arXiv exports); submitted and merged the fix upstream as [PR #4050](#).

SKILLS

Backend: Bun, Node.js, Express, Hono, Python, FastAPI, WebSockets, SSE, OAuth, JWT, PostgreSQL, MongoDB, Redis, Drizzle ORM, RabbitMQ, SQLAlchemy, Alembic

DevOps: Docker & Docker Compose, GitHub Actions (CI/CD), Git & GitHub, Nginx, AWS, Arch Linux (daily use)

AI/LLM: LangGraph, LangChain, RAG Pipelines, GraphRAG, Re-ranking RAG, Hybrid RAG, VectorDBs, RAGAS, Langfuse

Frontend: React, Next.js, JavaScript, TypeScript, Tailwind CSS, Shadcn/ui, Redux

EDUCATION

Software Engineering — MEng Equivalent (Diplôme National d'Ingénieur) Sep 2023 – Jul 2028

Higher Institute of Computer Science and Multimedia (ISIMS), Sfax, Tunisia

5-year integrated engineering program: 2-year preparatory cycle (Mathematics, Physics & Computer Science) + 3-year engineering cycle

CERTIFICATIONS · VIA COURSERA

IBM – RAG and Agentic AI (Jul 2026) · **AWS** – Cloud Solutions Architect (Feb 2026) · **U. Colorado Boulder** – Software Architecture for Big Data (Jan 2026)

LEADERSHIP & RECOGNITION

Software Engineering Department Lead — GDSC Enetcom 2026 – Present

- Lead the Software Engineering Department, designing a curriculum spanning backend development, system design, and AI engineering, and teaching it to the chapter.

Project Mentor — ISIMS Sfax 2025 – 2026

- Mentored a two-student team on their final-year graduation project (PFE), completed with high academic distinction.

National Student Entrepreneur Status (SNEE) — Initiator Level, University of Sfax 2025 – 2026

- Awarded for founding and operating a live commercial venture while completing an engineering degree.

LANGUAGES

English (Fluent) · French (Proficient) · Arabic (Native)